

now better understood, so that it is possible to be precise about dosages, aware of side effects, and confident in what form the herb should be taken as a medicine.

All over the world research is now being directed into plants with a traditional use as medicines, in the hope of finding new treatments for all manner of health problems. To give but two examples: the Indian gooseberry (*Emblica officinalis*, p. 204) appears to protect the liver against cancer, lowers blood cholesterol levels, and may prove valuable for acute pancreatitis; while thyme (*Thymus vulgaris*, p. 143), better known as a culinary herb, is a potent antioxidant that may prevent the breakdown of essential fatty acids in the brain and slow the aging process.

Medicinal Herbs & Big Business

The major pharmaceutical companies have realized that rainforests, grasslands, and even along roads and fields are sources of potentially invaluable medicines. As a result, the industry has invested vast resources into screening the active constituents of medicinal plants from all over the world. The drug taxol, first extracted from the Pacific yew (*Taxus brevifolia*), and one of the most effective treatments for breast cancer, was developed in this way.

In this search through the plant world it is likely that other remarkable medicines will be found, though far fewer new drugs have been successfully developed via this research than was originally anticipated. In fact, there is a key problem to this approach, for it is geared to the production of isolated plant chemicals which can then be synthesized and patented. With a patent, a company can make a profit, recouping the massive investment required to research and develop new medicines. Herbs, however, are whole, naturally occurring remedies. They cannot and should not be patented. Even if the major pharmaceutical companies were able to find an herb such as St. John's wort, which proved to be more effective and safer than conventional medicines, they would prefer to develop synthetic chemical drugs rather than plant medicines.

Herbal Synergy

One word more than any other separates herbal from conventional medicine: synergy. When the whole plant is used rather than extracted constituents, the different



St. John's wort's traditional use as a remedy for nervous exhaustion and depression has been confirmed by clinical trials.



The Moreton Bay chestnut is being investigated for its anti-HIV activity.

parts interact, often, it is thought, producing a greater therapeutic effect than the equivalent dosage of isolated active constituents that are generally preferred in conventional medicine.

Increasingly, research shows that herbs such as ephedra (*Ephedra sinica*, p. 95), hawthorn (*Crataegus oxyacantha*, p. 87), ginkgo (*Ginkgo biloba*, p. 100), and lily of the valley (*Convallaria majalis*, p. 194) have a greater-than-expected medicinal benefit thanks to the natural combination of constituents within the whole

plant. In some cases, the medicinal value of the herb may be due entirely to the combination of substances and cannot be reproduced by one or two "active" constituents alone.

The Future of Herbal Medicine

The main issue for the future of herbal medicine is whether medicinal plants, and the traditional knowledge that informs their use, will be valued for what they are—an immense resource of safe, economical, ecologically balanced medicines—or whether they will be yet another area of life to be exploited for short-term profit.

Another issue is convincing skeptics in the medical world that herbal medicine is not just a poor substitute for conventional medicine, but a valuable form of treatment in its own right. In trials into the effect of certain Chinese herbs on patients with eczema at London's Royal Free Hospital in the early 1990s, conventional specialists were astonished when the addition of one extra herb to a Chinese formula containing 10 others resulted in a dramatic improvement in a previously unresponsive patient. This story offers evidence of the skill and art involved in herbal practice. In tailoring the remedy to suit the individual needs of the patient and in treating the underlying cause, major improvements were made. This approach is a far cry from the standard medical view of using a single drug to treat a single disease.

In India and China, there have been university courses in herbal medicine for decades. In the West this process has been slower, with undergraduate courses emerging only in the last 20 years. There are now degree courses in herbal medicine or naturopathy in several Western countries, including Australia, the UK, and the U.S., while in Germany, medical students have been required to study naturopathy and phytotherapy (herbal medicine) as part of their medical training since 2003. Such developments point toward a future where patients might be able to choose between medical and herbal approaches when considering what medical treatment will suit them best.